

FAQ on Web Programming with Sample Servlet for Testing

Software for web programming

Please use the software version suggested under Homework 7 folder for the following exercises and practical test. (Tomcat v11.0; JDK 18; Eclipse IDE for Enterprise Java and Web Developers 2024-12)

Eclipse IDE for Enterprise Java and Web Developers 2024-12

https://www.eclipse.org/downloads/download.php?file=/technology/epp/downloads/release/2024-12/R/eclipse-jee-2024-12-R-win32-x86_64.zip

JDK (or JRE)

https://download.java.net/openjdk/jdk18/ri/openjdk-18+36_windows-x64_bin.zip

Tomcat 11

<https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.2/bin/apache-tomcat-11.0.2-windows-x64.zip>

MySQL connection

When you type the command, `mysqld`, to start the MySQL server, it should occupy the command prompt that you cannot type more command. If not, your MySQL server is still unable to start. Other applications are still unable to connect to the MySQL server as some applications may use the default port number requested by MySQL.

Please either restart your computer, uninstall any MySQL related software such as MySQL workbench or stop any Windows services with the name starting with “My” (Windows>Run>Type `services.msc`); Double click on the suspected service. Change Startup type to “Disabled” and click “Stop” at the Service status.

Please use the Class `TestConnection` provided in “Homework 7 Supplementary.pdf” under Homework 7 folder to test your MySQL server status and connection.

SQL

A `SELECT` statement can have an optional `WHERE` clause. The `WHERE` clause allows us to fetch records from a database table that matches specified condition(s). The result may be more than one row. For example,

```
-- select all columns from the customers table with last_name 'Doe'
```

```
SELECT * FROM Customers WHERE last_name = 'Doe';
```

You may play with the following online SQL editor:

<https://www.programiz.com/sql/online-compiler?preset=Customers1&ref=e4c06efc>

If the input criteria (`last_name` in the previous case) comes from input textbox or variable, remember to add the open and close quotation mark when making new SQL statement.

To test your SQL statement, you may open another command windows, type:

```
cd C:\temp\EIE3320\MySQL\mysql-8.3.0-winx64\bin
```

```
mysql -h localhost -u root
```

```
USE books;           --books is the name of Database created before for storing tables
```

```
SELECT * FROM BookInfo;
```

Eclipse Issue

If you see more than one red cross icon in your project, please stop and solve the issue first. If the problem cannot be resolved, please close Eclipse and delete or rename the whole eclipse-workspace. Then, create a new eclipse-workspace at C:\Temp\EIE3320 and create a new Dynamic Web Project. Please verify that the project is located within the directory C:\Temp\EIE3320\eclipse-workspace. Re-do homework 7 from scratch. As you generate a new Servlet, ensure that you retain all the default function headers generated. You should only transfer the essential portions of code into your new project. It is crucial to maintain consistency in class naming, ensuring that all classes are encapsulated within the same package. Avoid altering file names or class names after their creation. Additionally, do not modify the storage location for files managed by Eclipse's IDE.

If you cannot find the Servers Windows to change the port number, you may click: Windows>Show View>Other...>Type Server>Click Servers.

If Tomcat web server cannot be started, please change the Tomcat admin port from 8005 to another number or please delete the Tomcat Server at the Server Windows and create a new Server through the link to create a new server.

Should the Tomcat web server fail to launch, consider changing the administration port from its default of 8005 to another value. If this approach does not remedy the issue, you may need to delete the current Tomcat Server from the Servers panel in Eclipse and create a fresh server through the same Servers panel in Eclipse. You may restart your computer if the problem cannot be resolved.

Simple Sample Servlet for testing (able to run directly)

For Dynamic Web Applications in Eclipse, the IDE will generate project setting files for you. You cannot simply copy and paste the source code into a random file and expect it to run. You must provide the correct information (e.g., package name, class name, and file path) in the wizard. After that, you can copy and overwrite the source code. Alternatively, you can retain all generated import headers and function prototypes and overwrite the function content one by one. Therefore, if you see the following sample source code, you need to provide the following settings first:

The screenshot shows the 'Create Servlet' dialog box in Eclipse. The title bar says 'Create Servlet'. Below the title bar, it says 'Specify class file destination.' and there is a blue 'S' icon. The dialog has several input fields: 'Project' with a dropdown menu showing 'testing1', 'Source folder' with a text box containing '/testing1/src/main/java' and a 'Browse...' button, 'Java package' with a text box containing 'eie3320' and a 'Browse...' button, 'Class name' with a text box containing 'SimpleServlet' and a 'Browse...' button, and 'Superclass' with a text box containing 'jakarta.servlet.http.HttpServlet' and a 'Browse...' button. There is a checkbox labeled 'Use an existing Servlet class or JSP' which is unchecked. Below this checkbox is another 'Class name' field with 'SimpleServlet' and a 'Browse...' button. At the bottom, there are four buttons: a question mark icon, '< Back', 'Next >', and 'Finish' (which is highlighted with a blue border), and 'Cancel'.

```

package eie3320;

import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.PrintWriter;

/**
 * Servlet implementation class SimpleServlet
 */
@WebServlet("/SimpleServlet")
public class SimpleServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    /**
     * @see HttpServlet#HttpServlet()
     */
    public SimpleServlet() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
    response)
     */
    protected void doGet(HttpServletRequest request, HttpServletResponse
    response)
        throws ServletException, IOException {
        // TODO Auto-generated method stub
        response.getWriter().append("Served at:
    ").append(request.getContextPath());

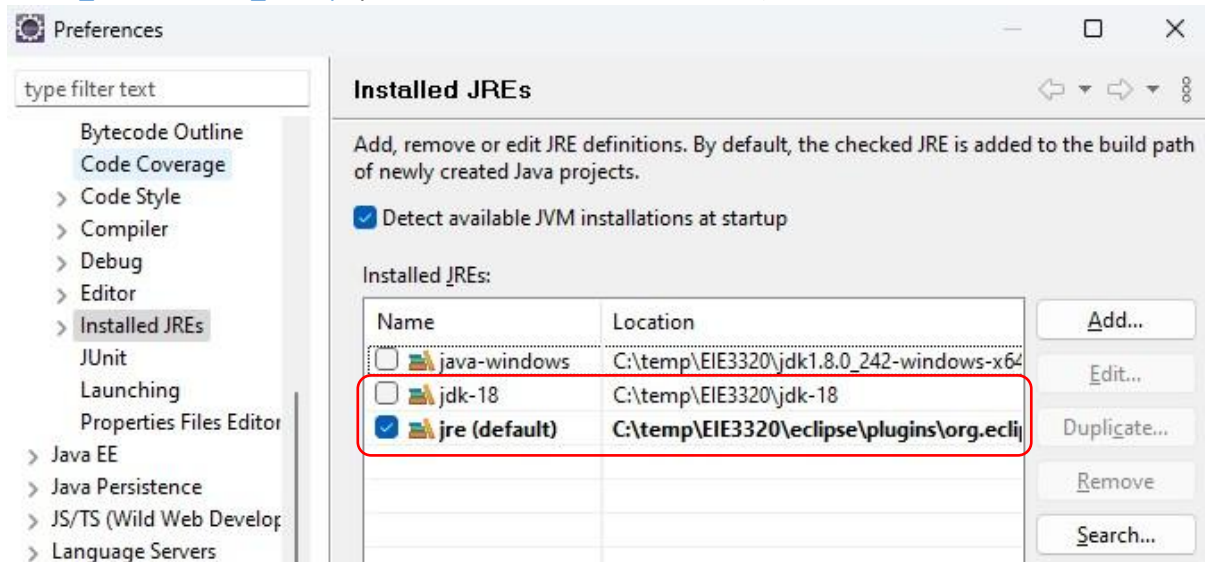
        PrintWriter out = response.getWriter();
        out.println("<br>Hello World!");
    }

    /**
     * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
    response)
     */
    protected void doPost(HttpServletRequest request, HttpServletResponse
    response)
        throws ServletException, IOException {
        // TODO Auto-generated method stub
        doGet(request, response);
    }
}

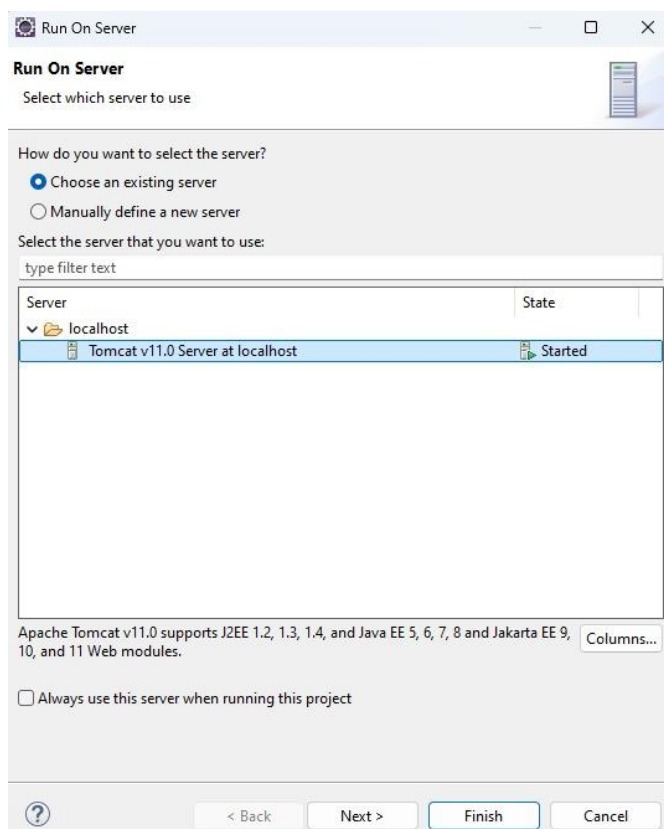
```

In case you cannot start the local tomcat server. Please check the following:

In Eclipse (2024-12), Windows > Preference > Installed JREs. Use either the default jre provided by Eclipse or the latest OpenJDK 18 (jdk-18 at https://download.java.net/openjdk/jdk18/ri/openjdk-18+36_windows-x64_bin.zip; posted also in Homework 7 folder)



Select available Server from the pull-down list or create . Do not click localhost.



Save all source codes. Remove old web server, restart your tomcat server or restart your computer if needed.

HTML

In SearchBook.html, the variable should look like the following: function

```
validateForm(){  
    var isbn = document.search.isbn.value;
```

If you do not mention the method, by default, it will call GET method. Please follow the instruction and do more practice on both “POST” and “Get” method.

```
<form method="POST" action="credit">
```

Servlet

The URL pattern is case-sensitive corresponding to Form action. e.g.

```
@WebServlet("/credit")  
public class CreditServlet extends HttpServlet {
```

```
<form method="POST" action="credit">
```